1. Biological Molecules

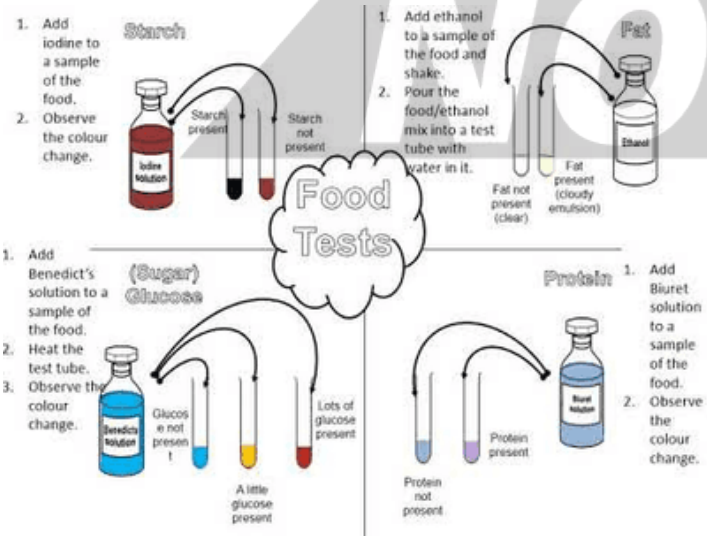
1.1. Biological Molecules

- **Carbohydrates:** made from Carbon, Hydrogen and Oxygen (CHO)
- **Fats and Oils:** made from Carbon, Hydrogen and Oxygen (CHO)
- **Proteins:** made from Carbon, Hydrogen, Oxygen, Nitrogen and sometimes Sulfur (CHON{S})

Smaller molecules	Larger molecules
Simple sugars	Starch, glycogen and cellulose
Fatty acids and glycerol	Fats and oils
Amino acids	Proteins

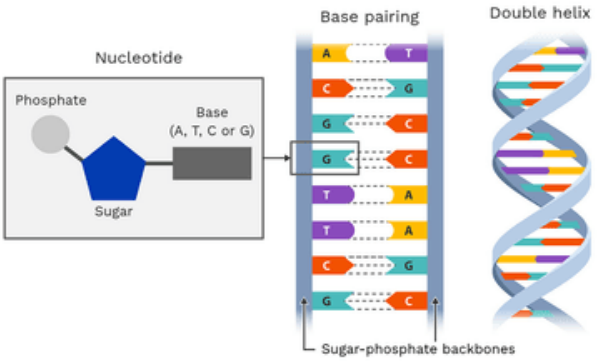
1.2. Food Tests

Test for:	Solution	Process	Positive Result	Negative Result
Reducing Sugars	Benedict Solution	Add Benedict solution into a beaker and Heat Up to 70-80°C	Brick Red	Blue
Starch	Iodine Solution	Add Iodine solution to the specimen	Blue-Black	Brown
Protein	Biuret Solution	Add Biuret solution to the sample	Purple/Violet	Blue
Fats (Physical test)	Ethanol	Add ethanol to the sample and shake with an equal volume of water.	Cloudy Emulsion	N/A
Vitamin C	DCPIP	Add DCPIP solution to the sample	Colourless	Blue



1.3. Structure of a DNA

- Chromosomes are made of a molecule called **DNA**
- DNA is also called **deoxyribonucleic acid**.



- Each chromosome is a very long molecule of tightly coiled DNA
- Two strands coiled together to form a **double helix**
- Each strand contains chemicals called **Bases**
- Names of DNA bases
- A stands for Adenine
- T stands for Thymine
- C stands for Cytosine
- G stands for Guanine

  (NOTE THAT THE FULL NAMES OF THE BASES ARE NOT REQUIRED FOR THE SYLLABUS)
- Cross-links between strands are formed by pairs of bases
- The bases always pair up in the same way, the base pairing rules are as follows:
  - **A and T**
  - **C and G**

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